

Conjecture 00000001025 is false:  
the extended binary QR code of length 24 is a second exception

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## Statement

**Conjecture 00000001025.** *The only exception to the statement that the automorphism group of the extended  $q$ -ary quadratic-residue code is a semidirect product of  $\text{PSL}(2, q)$  with commutative automorphisms is  $q = 7$  (a classification of QR-code automorphism groups).*

**Verdict.** False. For  $q = 23$  (the binary extended quadratic-residue code of length 24) the automorphism group is not a semidirect product, in either order, of  $\text{PSL}(2, 23)$  with a commutative group.

## Reading

The statement uses the letter  $q$  both for the alphabet (“ $q$ -ary”) and in  $\text{PSL}(2, q)$ , which acts on the  $p + 1$  coordinates of an extended QR code of prime length  $p$ . We refute two readings.

- **(R1)**  $q$  is the prime length parameter: for  $q = 23$  and the binary code (2 is a square mod 23), the automorphism group should be  $\text{PSL}(2, 23) \rtimes A$  or  $A \rtimes \text{PSL}(2, 23)$  with  $A$  commutative.
- **(R2)**  $q$  is the alphabet size: the same binary code ( $q = 2 \neq 7$ ) should have group  $\text{PSL}(2, 2) \rtimes A$  or  $A \rtimes \text{PSL}(2, 2)$ .

In both readings  $A$  is an arbitrary commutative group (so “commutative automorphisms” of any kind are covered), and the semidirect product may have either factor normal. The automorphism group of a binary code  $C \subseteq \mathbb{F}_2^{24}$  is the group of coordinate permutations  $\sigma$  with  $\sigma(C) = C$ ; for binary codes this equals the monomial automorphism group (the only nonzero scalar is 1) and there are no nontrivial field automorphisms of  $\mathbb{F}_2$ . Readings that use a different notion of “automorphism group” are not covered.

## The code

Coordinates are  $\mathbb{Z}/23 \cup \{\infty\}$  (in Lean: **Fin** 24, with 23 standing for  $\infty$ ). Let  $N$  be the set of quadratic non-residues mod 23 (the non-squares of  $\mathbb{Z}/23$ ). For  $t \in \mathbb{Z}/23$  let  $S_t = (t + N) \cup \{\infty\}$ , and let

$$W = \text{span}_{\mathbb{F}_2}(\{\mathbf{1}_{S_t} : t \in \mathbb{Z}/23\} \cup \{\mathbf{1}\}).$$

This is the construction quoted from Wikipedia (Binary Golay code): “Consider the set  $N$  of quadratic non-residues (mod 23) ... Augment each translate to a 12-element set  $S_t$  by adding an element  $\infty$  ...  $W$  can be defined as the span of the words  $S_t$  together with the word consisting of

all basis vectors”, and “the extended binary Golay code G24 consists of a 12-dimensional linear subspace  $W$ ”. Wikipedia (Quadratic residue code) lists “the (23,12) binary Golay code” among the quadratic residue codes, defines the QR code of length  $p$  over  $GF(l)$  by the generator polynomial  $\prod_{j \in Q}(x - \zeta^j)$ , and says that “adding an overall parity-check digit to a quadratic residue code gives an extended quadratic residue code”. The accompanying Python check (`check()` in the attempt file) builds  $GF(2^{11})$ , computes this generator polynomial for  $p = 23$ , appends the parity bit, and verifies that the resulting 4096-word code is *equal* to  $W$ . This identification is checked in Python only; the Lean file works with the  $S_t$  description above.

## Proof

**Lemma 1** (`card_aut_ge`).  $|\text{Aut}(W)| \geq 24 \cdot 23 \cdot 22 \cdot 21 \cdot 20 \cdot 16 = 81\,607\,680$ .

*Proof.* A systematic basis  $B_0, \dots, B_{11}$  of  $W$  on coordinates  $0, \dots, 11$  is given as bit masks; the kernel checks that every  $S_t$  and  $\mathbf{1}$  equals the combination of the  $B_k$  with its own coordinates  $0..11$  as coefficients, and that each  $B_k$  is an explicit combination of the  $S_t$  and  $\mathbf{1}$  (`W_eq`). Hence a permutation  $\sigma$  with  $B_k \circ \sigma = \sum_j (B_k \circ \sigma)(j) B_j$  for all  $k$  maps  $W$  into  $W$ , and,  $W$  being finite, onto  $W$  (`mem_aut`). Eleven explicit permutations (among them  $x \mapsto x + 1$  and  $x \mapsto -1/x$ ; the others were found by computer in the pointwise stabilizers) are certified this way (`gens_mem`). Let  $K_0 = \text{Aut}(W)$  and let  $K_{i+1}$  be the stabilizer in  $K_i$  of the base point  $a_i$ , with  $(a_0, \dots, a_5) = (\infty, 0, 1, 2, 3, 4)$ . For each  $i$ , the certified permutations fixing  $a_0, \dots, a_{i-1}$  move  $a_i$ , by explicit words, to 24, 23, 22, 21, 20, 16 distinct points respectively. If  $T$  is a set of points reachable from  $a$  in  $K$ , then  $(b, s) \mapsto g_b s$  is injective from  $T \times K_a$  to  $K$ , so  $|T| |K_a| \leq |K|$  (`card_le_of_orbit`, `card_le_of_words`). Chaining six steps gives the bound.  $\square$

(Consistently with this, Wikipedia (Mathieu group M24) states that  $M_{24}$ , of order 244,823,040, is the permutation automorphism group of  $W$ ; the proof uses only the lower bound.)

**Lemma 2** (`abelian_bound`, `card_comm_le`). *A commutative group  $A$  of permutations of 24 points has  $|A| \leq 3^8 = 6561$ .*

*Proof.* By strong induction on the size of a set  $T$  outside which  $A$  acts trivially: we show  $|A|^3 \leq 3^{|T|}$ . Pick  $x \in T$ , let  $O = Ax \subseteq T$  and  $A_x$  the stabilizer. Since  $A$  is commutative,  $A_x$  fixes  $O$  pointwise, so  $A_x$  acts trivially outside  $T \setminus O$ . The map  $h \mapsto (hx, r_{hx}^{-1}h)$  (with  $r_y \in A$ ,  $r_y x = y$ ) is injective from  $A$  to  $O \times A_x$ , so  $|A| \leq |O| |A_x|$ . With  $k^3 \leq 3^k$  for all  $k \in \mathbb{N}$  (`cube_le`) and the induction hypothesis,  $|A|^3 \leq 3^{|O|} 3^{|T \setminus O|} = 3^{|T|}$ . For  $T$  the whole set,  $|A|^3 \leq 3^{24}$ , so  $|A| \leq 3^8$ .  $\square$

**Lemma 3** (`card_SL_le`, `card_PSL_le`, `card_PSL2_le`).  $|\text{PSL}(2, \mathbb{Z}/23)| \leq |\text{SL}(2, \mathbb{Z}/23)| \leq 12144$  and  $|\text{PSL}(2, \mathbb{Z}/2)| \leq 16$  (the Lean statements record the common bound  $\leq 12144$ ).

*Proof.* Mathlib’s `Matrix.card_GL_field` gives  $|\text{GL}(2, 23)| = (23^2 - 1)(23^2 - 23) = 267168$ . The map  $(A, u) \mapsto \text{diag}(u, 1)A$  from  $\text{SL}(2, 23) \times (\mathbb{Z}/23)^\times$  to  $\text{GL}(2, 23)$  is injective (compare determinants, then cancel), so  $22 |\text{SL}(2, 23)| \leq 267168$ .  $\text{PSL}$  is a quotient of  $\text{SL}$  (Mathlib:  $\text{SL}$  modulo its center).  $\text{SL}(2, \mathbb{Z}/2)$  consists of  $2 \times 2$  matrices over  $\mathbb{F}_2$ , at most 16.  $\square$

**Theorem 4** (`not_semidirect_left`, `not_semidirect_right`, `conjecture_1025_false`). *For every group  $N$  with  $|N| \leq 12144$  and every commutative group  $A$  there is no group isomorphism  $\text{Aut}(W) \cong N \rtimes_\varphi A$  or  $\text{Aut}(W) \cong A \rtimes_\varphi N$ . In particular this holds for  $N = \text{PSL}(2, \mathbb{Z}/23)$  and  $N = \text{PSL}(2, \mathbb{Z}/2)$ .*

*Proof.* An isomorphism gives  $|\text{Aut}(W)| = |N| |A|$  (Mathlib `SemidirectProduct.card`), and  $A$  embeds (via `inr` resp. `inl`) into  $\text{Aut}(W) \leq \text{Sym}(24)$  as a commutative subgroup, so  $|A| \leq 6561$ . Then  $|\text{Aut}(W)| \leq 12144 \cdot 6561 = 79\,676\,784 < 81\,607\,680$ , a contradiction.  $\square$

Hence  $q = 23$  (reading R1) and  $q = 2$  (reading R2) are exceptions different from  $q = 7$ , and the conjecture is false.

## Lean formalization

File `Conjecture1025/Basic.lean` (namespace `C1025`), only import `Mathlib`. Main theorem `C1025.conjecture_1025_false`: the four statements  $\forall A [\text{CommGroup } A] \forall \varphi, \text{IsEmpty}(\text{Aut } W \simeq^* N \rtimes_{\varphi} A)$  and  $\text{IsEmpty}(\text{Aut } W \simeq^* A \rtimes_{\varphi} N)$  for  $N = \text{PSL}(2, \mathbb{Z}/23\mathbb{Z})$  and  $N = \text{PSL}(2, \mathbb{Z}/2\mathbb{Z})$ . Definitions: `S`, `one24`, `W` (span of the  $S_t$  and the all-ones word), `Aut` (the subgroup  $\{\sigma : \forall v, v \in C \leftrightarrow v \circ \sigma \in C\}$  of `Equiv.Perm (Fin 24)`). Quadratic non-residue is `~IsSquare` in `ZMod 23` (0 is a square, so non-residues are nonzero). All finite computations use `decide +kernel`; there is no `native_decide` and no `sorry`.

**Axiom audit.** `#print axioms C1025.conjecture_1025_false reports [propext, Classical.choice, Quot.sound]`.

## Sources

- [https://en.wikipedia.org/wiki/Quadratic\\_residue\\_code](https://en.wikipedia.org/wiki/Quadratic_residue_code) (definition, generator polynomial, extended code, examples).
- [https://en.wikipedia.org/wiki/Binary\\_Golay\\_code](https://en.wikipedia.org/wiki/Binary_Golay_code) (construction of  $W$  from the  $S_t$ ).
- [https://en.wikipedia.org/wiki/Mathieu\\_group\\_M24](https://en.wikipedia.org/wiki/Mathieu_group_M24) (consistency only).